



2008

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start a new booklet for each question

Total marks – 120

- Attempt Questions 1 – 8
- All questions are of equal value

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Total Marks – 120

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) $\int \frac{2x}{\sqrt{1-x^4}} dx$ 2

(b) $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$ 3

(c) $\int_1^{e^2} 3x^2 \ln x \, dx$ 3

(d) $\int \frac{dx}{\sqrt{x^2 - x + 1}}$ 2

(e) (i) Show that $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$ 2

(ii) Use this property to show that $\int_0^1 x^3(1-x)^6 \, dx = \frac{1}{840}$ 3

Question 2 (15 marks) Use a SEPARATE writing booklet.

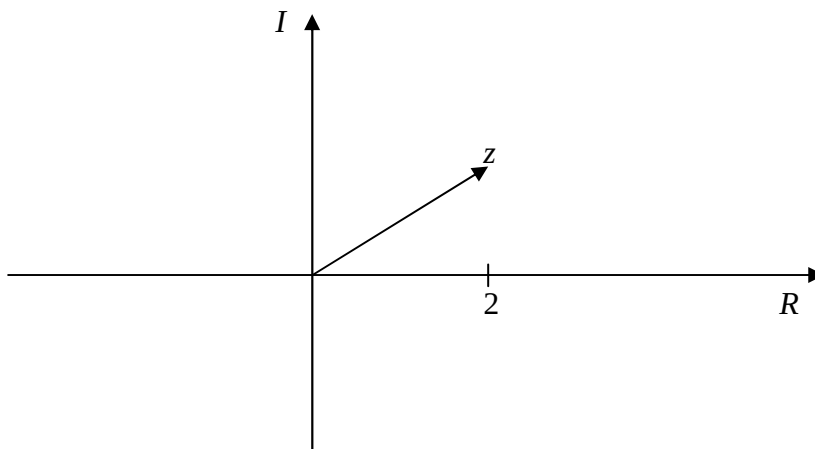
Marks

(a) A complex number z is given by $z = \sqrt{3} + i$

(i) Evaluate $\bar{z}$. Verify that $z\bar{z}$ is real. 2

(ii) Find $\frac{1}{z}$ in the form $a + ib$, where a and b are real. 1

(b) A point z on the Argand Diagram is given below:



Copy this diagram into your examination booklet and use it to plot the following points. Clearly label each point.

(i) $\bar{z}$ 1

(ii) $2iz$ 1

(iii) $\frac{1}{z}$ 1

(c) Express $i - 1$ in modulus argument form, and hence simplify $(i - 1)^5$ 2

Question 2 continues on page 4

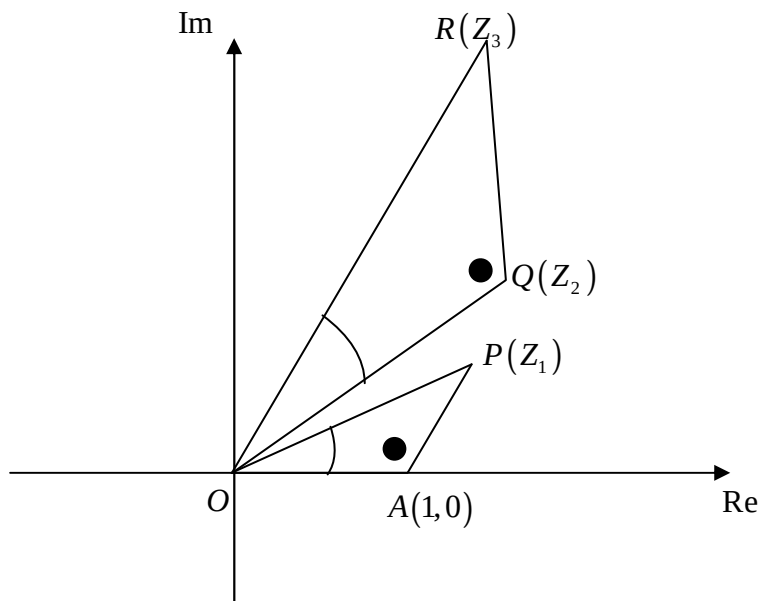
Question 2 (continued)

(d) Sketch the locus and state its equation:

(i) $|z - 2| = |z - 2i|$ 2

(ii) $z\bar{z} - 3(z + \bar{z}) \leq 0$ 2

(e)



In the figure above, the points P , Q and A represent the complex numbers Z_1, Z_2 and $(1,0)$ respectively. Given $\angle OAP = \angle OQR$ and $\angle AOP = \angle QOR$.

Explain why $R(Z_3)$ represents the complex number $Z_1 Z_2$. 3

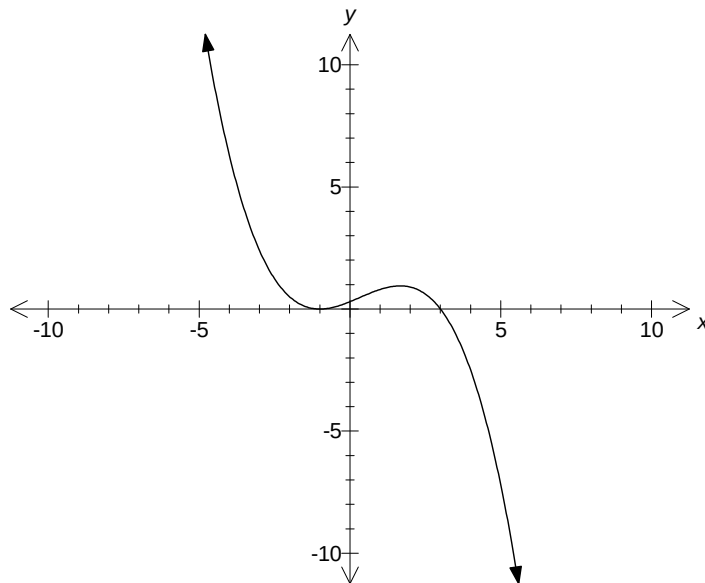
You must support your answer with clear and complete mathematical reasons.

End of Question 2

Question 3 (15 marks) Use a SEPARATE writing booklet.

Marks

(a)



The graph of $f(x) = \frac{1}{10}(x+1)^2(3-x)$ is drawn above.

On separate diagrams, draw a neat sketch showing the main features of each of the following:

- | | | |
|-------|------------------|----------|
| (i) | $y = f(x-1)$ | 1 |
| (ii) | $y = f(x)$ | 1 |
| (iii) | $y = \{f(x)\}^2$ | 2 |
| (iv) | $y = xf(x)$ | 2 |
| (v) | $y^2 = f(x)$ | 2 |
| (vi) | $y = e^{f(x)}$ | 2 |

(b) Given that $I_n = \int_0^{\frac{\pi}{4}} \sec^n x \, dx$ show that :

- | | | |
|------|--|----------|
| (i) | $I_n = \frac{1}{n-1} \left((\sqrt{2})^{n-2} + (n-2)I_{n-2} \right)$ | 4 |
| (ii) | Hence or otherwise evaluate I_4 | 1 |

End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) (i) If $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$, show that $P(x) = 0$ has a multiple root, find this root and its multiplicity. **3**
- (ii) Hence factorise $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$ into its linear factors. **1**
- (b) The equation $x^3 + 2x - 1 = 0$ has roots α, β, γ . Find the monic equations with roots
- (i) $\alpha^2, \beta^2, \gamma^2$. **2**
- (ii) $\alpha\beta, \beta\gamma, \alpha\gamma$ **3**
- (iii) Evaluate $\alpha^3 + \beta^3 + \gamma^3$ **2**
- (c) A point $P\left(ct, \frac{c}{t}\right)$ lies on the rectangular hyperbola $xy = c^2$.
- (i) Show that the equation of the tangent at the point $P\left(ct, \frac{c}{t}\right)$ on the rectangular hyperbola is given by $x + t^2y = 2ct$. **2**
- (ii) Prove that the area bounded by the tangent and the asymptotes of the rectangular hyperbola is a constant. **2**

End of Question 4

Question 5 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) ABC is an equilateral triangle, inscribed in a circle. X is a point on the minor arc BC .

(i) Prove that $\triangle BD X \parallel \triangle AC X$

3

(iii) Prove that $XB + XC = XA$

3

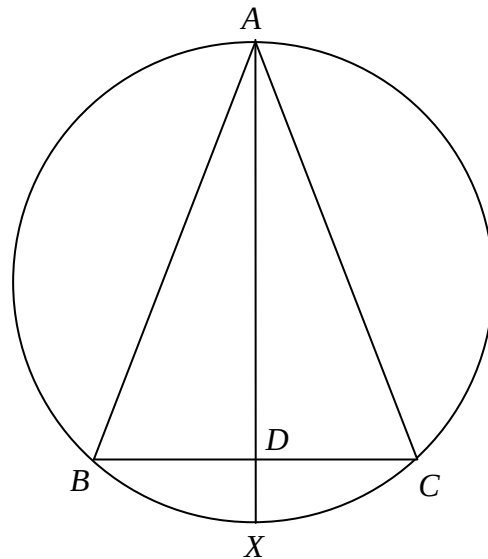


DIAGRAM NOT TO SCALE

(b) State whether each of the following are true or false giving brief reasons for your answers:

(i) $\int_0^{\pi} \sin 9x \, dx = 0$

1

(ii) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \sin x \, dx = 0$

2

(c) Find the equation of the tangent to the curve $\cos 2x + \sin y = 1$ at the point $x = \frac{\pi}{6}$.

3

(d) Use the substitution $x = a \sin \theta$ to show that

3

$$\int \sqrt{a^2 - x^2} \, dx = \frac{1}{2} a^2 \sin^{-1} \frac{x}{a} + \frac{1}{2} x \sqrt{a^2 - x^2} + C$$

End of Question 5

Question 6 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) (i) Given $a\alpha^2 + b\alpha + c = 0$ where $a, b, c \in \mathbb{R}$ and $\alpha \in \mathbb{C}$, prove that $a(\overline{\alpha})^2 + b\overline{\alpha} + c = 0$ 2
- (ii) A polynomial $P(x)$ with real coefficients, has two of its zeros $3i$ and $1 + 2i$. Find in expanded form, a possible polynomial $P(x)$. 3
- (b) Use De Moivre's Theorem and binomial expansion to find an expression for $\cos 4\theta$ in terms of $\cos \theta$. 3
- (c) (i) Given $z = \cos \theta + i \sin \theta$, prove $z^n + \frac{1}{z^n} = 2 \cos n\theta$ 2
- (i) Hence by considering the expansion $\left(z + \frac{1}{z}\right)^4$ show that 3
- $$\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$$
- (iii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^4 \theta \, d\theta$ 2

End of Question 6

Question 7 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The roots of the polynomial $p(x) = x^3 + ax^2 + bx + c = 0$ are three consecutive terms of an arithmetic series. Prove that the relationship between the coefficients is given by $2a^3 - 9ab + 27c = 0$

4

Hint: make an appropriate choice for the roots in arithmetic progression.

- (b) A point $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a > 0$ and $b > 0$.

The equation of the normal at the point $P(a \cos \theta, b \sin \theta)$ is given by

$$xa \sin \theta - yb \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

- (i) Show that the ellipse intersects the rectangular hyperbola $xy = c^2$ in four points if $ab > 2c^2$

3

- (ii) Show that for $0 < \theta < \frac{\pi}{2}$, the normal at P on the ellipse intersects the hyperbola in two distinct points, say A and B .

3

- (iii) If M is the mid-point of AB , show that the coordinates of M are given by

2

$$\left(\frac{(a^2 - b^2) \cos \theta}{2a}, -\frac{(a^2 - b^2) \sin \theta}{2b} \right)$$

- (iv) Hence find the locus of M as θ varies.

3

End of Question 7

Question 8 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) For the function $y = \cos^{-1}(e^x)$,
- (i) Find the domain and the range. 2
 - (ii) Draw a neat sketch the graph of $y = \cos^{-1}(e^x)$. 2
 - (iii) Hence draw a neat sketch of the curve $y = \frac{1}{(\cos^{-1}(e^x))}$ 2
- (b) (i) Using induction, show that for each positive integer n , there are unique positive integers p_n and q_n such that: $(1+\sqrt{2})^n = p_n + q_n\sqrt{2}$ 4
- (ii) Show also that $p_n^2 - 2q_n^2 = (-1)^n$. 1
- (c) If $f(xy) = f(x) + f(y)$, for all $x, y \neq 0$, prove that
- (i) $f(1) = f(-1) = 0$ 2
 - (ii) $f(x)$ is an even function. 2

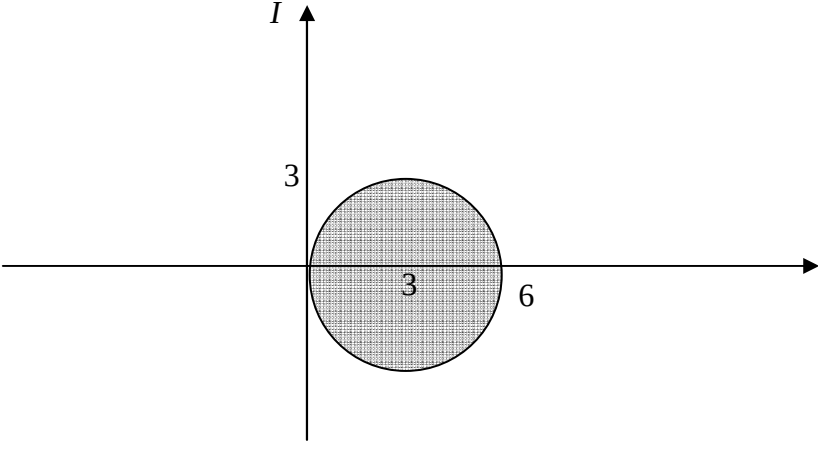
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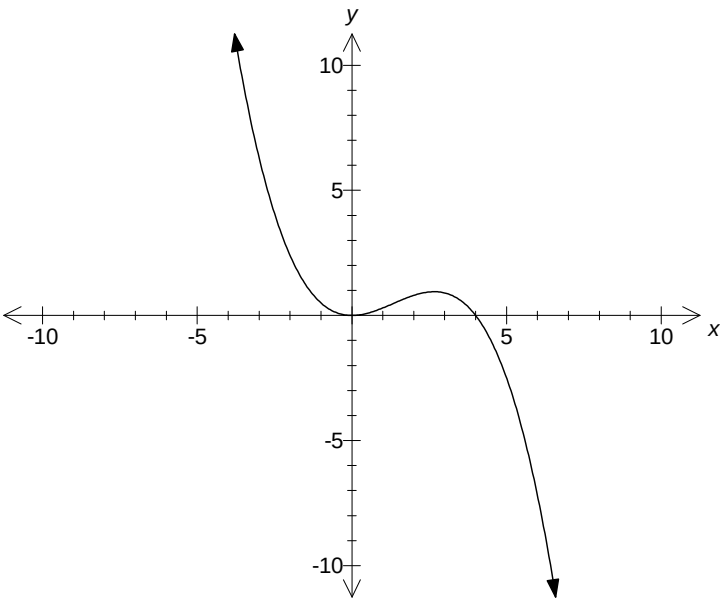
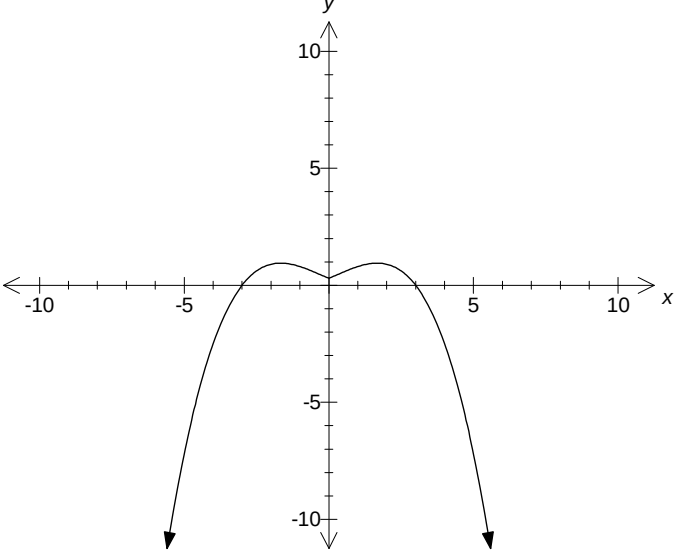
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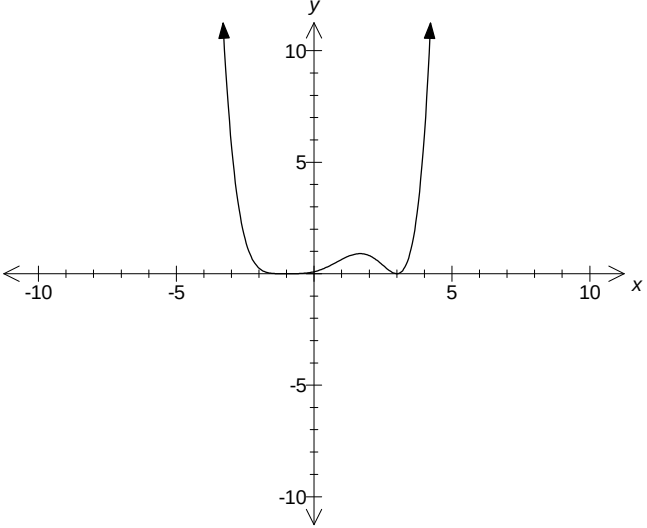
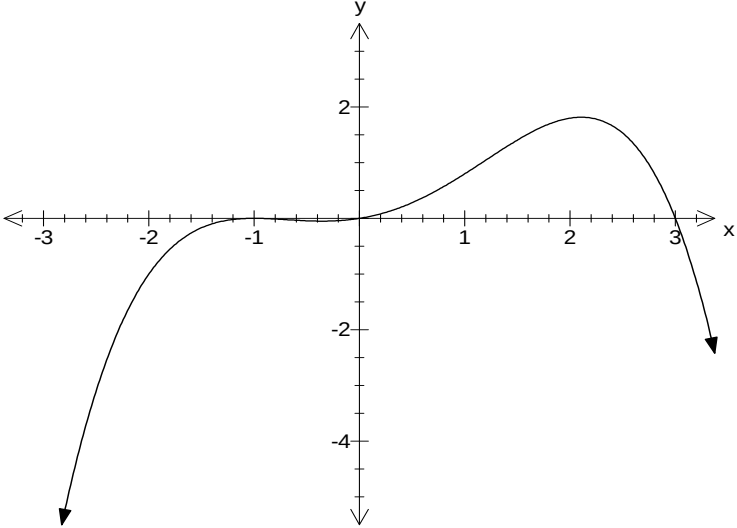
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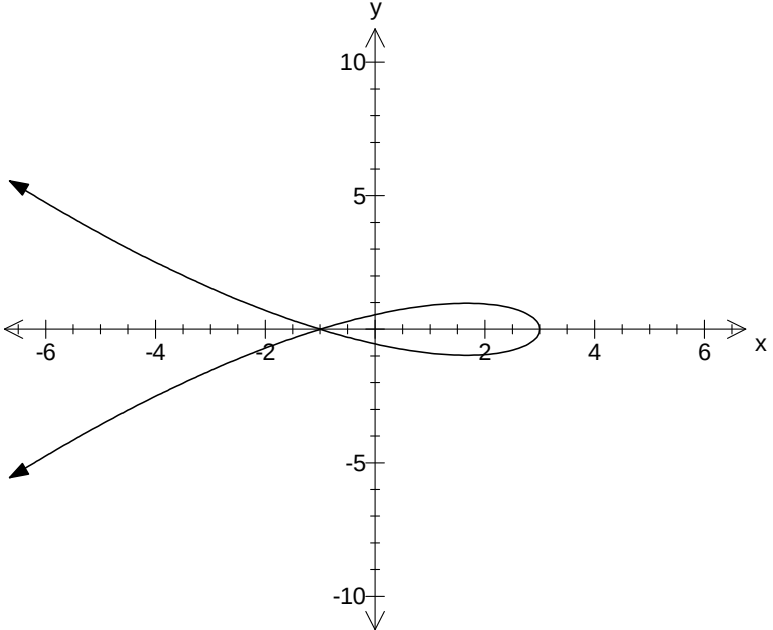
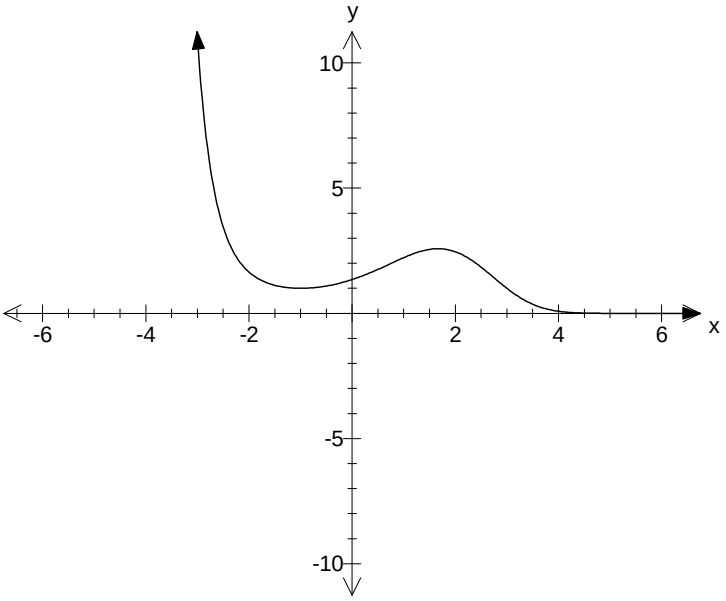
Question	Criteria	Marks	Bands
1(a)	$\int \frac{2x}{\sqrt{1-x^4}} dx \quad \text{Let } u = x^2 \quad \therefore \frac{du}{dx} = 2x \quad \text{or} \quad dx = \frac{du}{2x} \quad \boxed{\checkmark}$ $\therefore \int \frac{2x}{\sqrt{1-x^4}} dx = \int \frac{2x}{\sqrt{1-u^2}} \cdot \frac{du}{2x}$ $= \int \frac{1}{\sqrt{1-u^2}} du$ $= \sin^{-1} u + C$ $= \sin^{-1} x^2 + C \quad \boxed{\checkmark}$	2	
1(b)	$\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$ $\text{Let } t = \tan \frac{x}{2}$ $\therefore \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2} = \frac{1}{2} [1 + \tan^2 \frac{x}{2}] = \frac{1}{2} [1 + t^2] \quad \text{or} \quad dx = \frac{2 dt}{1+t^2}$ $\text{and } \cos \theta = \frac{1-t^2}{1+t^2} \quad \boxed{\checkmark}$ $\therefore \int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x} = \int_0^1 \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{2 dt}{1+t^2}$ $\text{since } t = \tan \frac{\frac{\pi}{2}}{2} = 1 \quad \text{and} \quad t = \tan \frac{0}{2} = 0$ $= \int_0^1 \frac{1}{\frac{2(1+t^2)}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2 dt}{1+t^2}$ $= \int_0^1 \frac{1+t^2}{3+t^2} \cdot \frac{2 dt}{1+t^2} \quad \boxed{\checkmark}$ $= \int_0^1 \frac{2}{3+t^2} dt$ $= 2 \int_0^1 \frac{1}{3+t^2} dt$ $= 2 \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1 \quad \boxed{\checkmark}$ $= \frac{\pi}{3\sqrt{3}}$	3	

1(c)	$\int_1^{e^2} 3x^2 \ln x \, dx \quad \text{Let } u = \ln x \quad \frac{dv}{dx} = 3x^2 \quad \frac{du}{dx} = \frac{1}{x} \quad v = x^3$ $\therefore \int_1^{e^2} 3x^2 \ln x \, dx = uv - \int v \, du \quad \checkmark$ $= \left[x^3 \ln x \right]_1^{e^2} - \int x^3 \frac{dx}{x}$ $= \left[x^3 \ln x \right]_1^{e^2} - \int x^2 \, dx$ $= \left[x^3 \ln x \right]_1^{e^2} - \left[\frac{x^3}{3} \right]_1^{e^2} \quad \checkmark$ $= \left[e^6 \ln e^2 - 1^3 \ln 1 \right] - \left[\frac{e^6}{3} - \frac{1}{3} \right]$ $= 2e^6 - \frac{e^6}{3} + \frac{1}{3}$ $= \frac{5e^6 + 1}{3} \quad \checkmark$	3	
1(d)	$\int \frac{dx}{\sqrt{x^2 - x} + 1} = \int \frac{1}{\sqrt{x^2 - x + (\frac{1}{2})^2 + 1 - (\frac{1}{2})^2}} \, dx \quad \checkmark$ $= \int \frac{1}{\sqrt{(x - \frac{1}{2})^2 + \frac{3}{4}}} \, dx$ $= \log \left (x - \frac{1}{2}) + \sqrt{(x - \frac{1}{2})^2 + \frac{3}{4}} \right + C \quad \checkmark$	2	
1(e)(i)	$\int_0^a f(a-x) \, dx \quad \text{Let } u = a-x \quad \therefore \frac{du}{dx} = -1$ $\text{If } x=a \text{ then } u=a-a=0 \quad x=0 \text{ then } u=a-0=a \quad \checkmark$ $\int_0^a f(u) \cdot -du = - \int_a^0 f(u) \, du$ $= \int_0^a f(u) \, du \quad \checkmark$	2	
1(e)(ii)	$\int_0^1 x^3 (1-x)^6 \, dx = \int_0^1 (1-x)^3 (1-(1-x))^6 \, dx \quad \checkmark$ $= \int_0^1 (1-3x+3x^2-x^3)x^6 \, dx$ $= \int_0^1 x^6 - 3x^7 + 3x^8 - x^9 \, dx \quad \checkmark$ $= \left[\frac{x^7}{7} - \frac{3x^8}{8} + \frac{x^9}{3} - \frac{x^{10}}{10} \right]_0^1 \quad \checkmark$ $= \frac{1}{840}$	3	

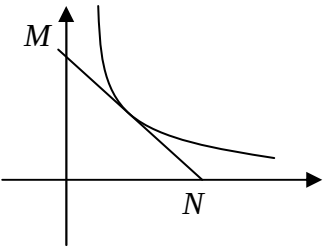
2(d)(ii)	 $(x-3)^2 + y^2 \leq 9$	1	
2(e)	$\angle AOR = \angle AOQ + \angle QOR$ $= \angle AOQ + \angle AOP$ <i>i.e.</i> $\arg z_3 = \arg z_2 + \arg z_1$ the triangles ORQ and OPA are equiangular and hence similar $\therefore$ their sides are proportional $\frac{OR}{OP} = \frac{OQ}{OA} = \frac{ z_3 }{ z_1 } = \frac{ z_2 }{ 1 }$ $z_3 = \frac{ z_2 z_1 }{1}$ $= z_2 z_1$ $\therefore z_3 = z_2 \cdot z_1$ <i>i.e.</i> R represents the complex number $z_2 z_1$	1 1 1	

Question	Criteria	Marks	Bands
3(a)(i)	 ☒ <i>shift graph 1 place to the right</i>	1	
3(a)(ii)	 ☒ <i>reflection in y axis</i>	1	

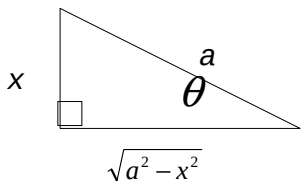
3(a)(iii)	 ☒ graph above x-axis ☒ graph $0 < y < 1$ (less steep gradient) and graph $y > 1$ steeper gradient	2	
3(a)(iv)	 ☒ 3 roots at -1, 0 and 3 ☒ 2 stationary points	2	

3(a)(v)	 ☒ sketch only + ve section of original graph and reflect in x - axis ☒ for $-1 < y < 1$ the sketch is less steep and more steep after $y < -1$ and $y > 1$	2	
3(a)(vi)	 ☒ where cuts x - axis now becomes $(-1,1)$ and $(3,1)$ ☒ anything that was negative now becomes an asymptote above the x - axis.	2	

3(b)(i)	$I_n = \int_0^{\frac{\pi}{4}} \sec^n x \, dx$ $\int \sec^n x \, dx = \int \sec^{n-2} x \cdot \sec^2 x \, dx \quad \checkmark$ where $u = \sec^{n-2} x \quad \frac{dv}{dx} = \sec^2 x$ $\frac{du}{dx} = (n-2)\sec^{n-3} x \sec x \tan x \quad v = \tan x$ $\int \sec^{n-2} x \cdot \sec^2 x \, dx = uv - \int v \, du$ $= \sec^{n-2} x \tan x - \int (n-2)\sec^{n-3} x \sec x \tan^2 x \, dx$ $= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x \tan^2 x \, dx$ $= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) \, dx$ $= \sec^{n-2} x \tan x - (n-2) \int \sec^n x - \sec^{n-2} x \, dx \quad \checkmark$ $\therefore \int \sec^n x \, dx = \sec^{n-2} x \tan x - (n-2) \int \sec^n x - (n-2) \int \sec^{n-2} x \, dx$ $\int \sec^n x \, dx + (n-2) \int \sec^n x = \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x \, dx$ $(n-1) \int \sec^n x \, dx = \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x \, dx$ $\int \sec^n x \, dx = \frac{1}{n-1} \sec^{n-2} x \tan x - \frac{n-2}{n-1} \int \sec^{n-2} x \, dx \quad \checkmark$ $I_n = \frac{1}{n-1} \left[\left[\sec^{n-2} x \tan x \right]_0^{\frac{\pi}{4}} - (n-2) I_{n-2} \right]$ $I_n = \frac{1}{n-1} \left[\left[\sec^{n-2} \left(\frac{\pi}{4} \right) \tan \left(\frac{\pi}{4} \right) - \sec^{n-2} (0) \tan(0) \right] - (n-2) I_{n-2} \right] \quad \checkmark$ $I_n = \frac{1}{n-1} \left[\left(\sqrt{2} \right)^{n-2} - (n-2) I_{n-2} \right]$	4	
3(b)(ii)	$I_4 = \frac{1}{3} \left[\left(\sqrt{2} \right)^2 - 2I_2 \right]$ $I_2 = \frac{1}{2} \left[\left(\sqrt{2} \right)^0 - 0I_0 \right] = \frac{1}{2}$ $\therefore I_4 = \frac{1}{3} \left[\left(\sqrt{2} \right)^2 - 2 \left(\frac{1}{2} \right) \right]$ $= \frac{2}{3} - \frac{1}{3}$ $= \frac{1}{3} \quad \checkmark$	1	

Question	Criteria	Marks	Bands
4(a)(i)	$P'(x) = 4x^3 + 3x^2 - 6x - 5$ $P''(x) = 12x^2 + 6x - 6$ $12x^2 + 6x - 6 = 0 \Rightarrow x = -1, -\frac{1}{2}$ $P'(-1) = 0$ $\therefore x = -1$ is a root of multiplicity 3	1 1 1	
4(a)(ii)	$x^4 + x^3 - 3x^2 - 5x - 2 = (x+1)^3(x-2)$	1	
4(b)(i)	$(\sqrt{x})^3 + 2\sqrt{x} - 1 = 0$ $x\sqrt{x} + 2\sqrt{x} = 1$ $(\sqrt{x}(x+2))^2 = 1$ $x(x^2 + 4x + 4) = 1$ $x^3 + 4x^2 + 4x - 1 = 0$	1	
4(b)(ii)	$\alpha\beta, \alpha\gamma, \beta\gamma = \frac{\alpha\beta\gamma}{\gamma}, \frac{\alpha\beta\gamma}{\beta}, \frac{\alpha\beta\gamma}{\alpha} = \frac{1}{\gamma}, \frac{1}{\beta}, \frac{1}{\alpha}$ $\left(\frac{1}{x}\right)^3 + 2\left(\frac{1}{x}\right) - 1 = 0$ $\frac{1}{x^3} + \frac{2}{x} - 1 = 0$ $x^3 - 2x^2 - 1 = 0$	1 1 1	
4(b)(iii)	$\alpha^3 + 2\alpha - 1 = 0$ $\beta^3 + 2\beta - 1 = 0$ $\gamma^3 + 2\gamma - 1 = 0$ $(\alpha^3 + \beta^3 + \gamma^3) + 2(\alpha + \beta + \gamma) - 3 = 0$ $\therefore \alpha^3 + \beta^3 + \gamma^3 = 3 - 2(0)$ $= 3$	1 1	
4(c)(i)	$y' = -\frac{c}{x^2}$ at $\left(ct, \frac{c}{t}\right)$ $y' = -\frac{1}{t^2}$ $y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$ $yt^2 - ct = -x + ct$ $yt^2 + x = 2ct$	1 1	
4(c)(ii)	$M : x = 0 \Rightarrow y = \frac{2c}{t}$ $N : y = 0 \Rightarrow x = 2ct$ $\text{area} = \frac{1}{2}(2ct)\left(\frac{2c}{t}\right)$ $= 2c^2$ which is a constant	 1 1	

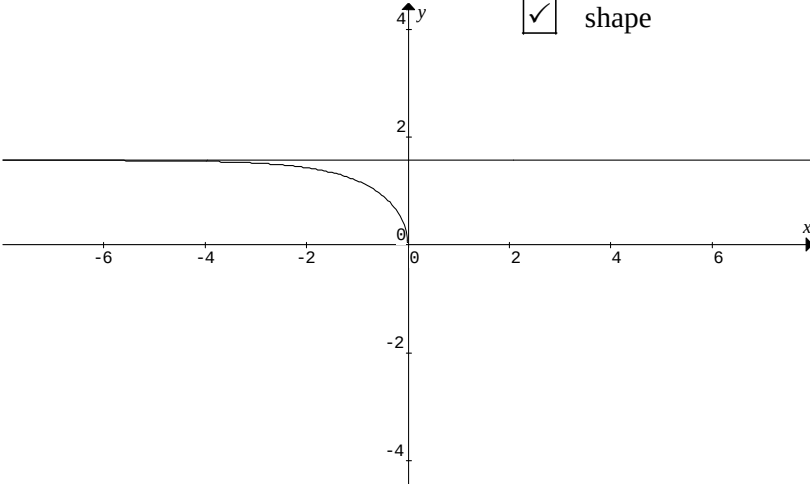
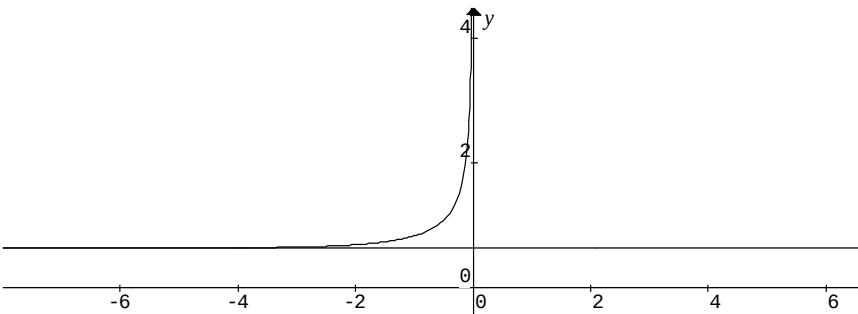
Question	Criteria	Marks	Bands
5(a)(i)	$\angle BAC = 60$ ($\triangle ABC$ is equilateral $\triangle$) $\angle BXC = 120$ (opposite angles in cyclic quad ABXC are supplementary) $\angle AXC = \angle ABC = 60$ ($\angle$'s on same arc at circumference are equal) $\therefore \angle AXC = \angle AXB = 60$ in $\triangle BDX$ and $\triangle ACX$ $\angle DXC = \angle AXB = 60$ (proved above) ☒ $\angle DXB = \angle CAX$ ($\angle$'s at circumference on same arc) ☒ $\therefore \triangle BDX \parallel \triangle ACX$ (equiangular) ☒	3	
5(a)(ii)	$\triangle CDX \parallel \triangle ABX$ (as proved in (i) above) since $\triangle BDX \parallel \triangle ACX$ $\therefore \frac{BD}{AC} = \frac{BX}{AX} = \frac{DX}{CX}$ and $\frac{CD}{AB} = \frac{CX}{AX} = \frac{DX}{BX}$ $\therefore BD = \frac{BX \cdot AC}{AX}$ and $CD = \frac{CX \cdot AB}{AX}$ ☒ since $BC = BD + DC$ hence $BC = \frac{BX \cdot AC}{AX} + \frac{CX \cdot AB}{AX}$ ☒ $BC \cdot AX = (BX \cdot AC) + (CX \cdot AB)$ and as $BC = AC = AB$ (equilateral $\triangle$) $\therefore \div$ LHS and RHS by BC ☒ $\therefore AX = BX + CX$	3	
5(b)(i)	$y = \sin 9x$ when $0 < x < \pi$ it has $4\frac{1}{2}$ cycles, more area above x -axis than below $\therefore \int_0^{\pi} \sin 9x \, dx \neq 0$ (false) ☒	1	
5(b)(ii)	$x \sin x$ is an even function ☒ $\therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \sin x \, dx = 2 \int_0^{\frac{\pi}{2}} x \sin x \, dx \neq 0$ ☒ (false)	2	

5(c)	slope tangent = $\frac{dy}{dx}$: (differentiating implicitly) $-2 \sin 2x + \cos y \frac{dy}{dx} = 0$ $\cos y \frac{dy}{dx} = 2 \sin 2x$ $\frac{dy}{dx} = \frac{2 \sin 2x}{\cos y} \quad \boxed{\checkmark}$ At $(\frac{\pi}{6}, \frac{\pi}{6})$, slope of tangent = 2 $\boxed{\checkmark}$ Equation of tangent is $y - \frac{\pi}{6} = 2(x - \frac{\pi}{6})$ $y = 2x - \frac{\pi}{6} \quad \boxed{\checkmark}$	3	
5(d)	Let $x = a \sin \theta \quad dx = a \cos \theta d\theta$ $\int \sqrt{a^2 - x^2} dx = \int \sqrt{a^2 - a^2 \sin^2 \theta} a \cos \theta d\theta = a^2 \int \cos^2 \theta d\theta \quad \boxed{\checkmark}$ $= \frac{a^2}{2} \int \cos 2\theta + 1 d\theta =$ $\frac{a^2}{2} \left[\frac{1}{2} \sin 2\theta + \theta \right] + C = \frac{a^2}{2} [\sin \theta \cos \theta + \theta] + C \quad \boxed{\checkmark}$ From this triangle :  $\frac{x}{a} = \sin \theta$ $\cos \theta = \frac{\sqrt{a^2 - x^2}}{a}$ $\theta = \sin^{-1} \frac{x}{a}$ $\frac{a^2}{2} [\sin \theta \cos \theta + \theta] + C =$ $\frac{a^2}{2} \left[\frac{x}{a} \frac{\sqrt{a^2 - x^2}}{a} + \sin^{-1} \frac{x}{a} \right] + C = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C \quad \boxed{\checkmark}$	3	

Question	Criteria	Marks	Bands
6(a)(i)	$\overline{a\alpha^2 + b\alpha + c} = \overline{0}$ $\overline{a\alpha^2 + b\alpha + c} = 0$ $a\overline{\alpha^2} + b\overline{\alpha} + \overline{c} = 0$ $a\overline{\alpha}^2 + b\overline{\alpha} + \overline{c} = 0$ $\therefore \overline{\alpha}$ is a solution	1 1	
6(a)(ii)	$(x - 3i)(x + 3i)(x - (1 + 2i))(x - (1 - 2i))$ $(x^2 + 9)(x^2 - 2x + 5)$ $x^4 - 2x^3 + 14x^2 - 18x + 45$	1 1 1	
6(b)	$\cos 4\theta + i \sin 4\theta = (\cos \theta + i \sin \theta)^4$ $= \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta$ equate real part: $\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$ $= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$ $= 8 \cos^4 \theta - 8 \cos^2 \theta + 1$	1 1 1	
6(c)(i)	$z^n = \cos n\theta + i \sin n\theta$ $z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$ $= \cos n\theta - i \sin n\theta$ $z^n + z^{-n} = \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$ $= 2 \cos n\theta$	1 1	
6(c)(ii)	$\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$ $= \left(z^4 + \frac{1}{z^4}\right) + 4\left(z^2 + \frac{1}{z^2}\right) + 6$ $= 2 \cos 4\theta + 4(2 \cos 2\theta) + 6$ $\left(z + \frac{1}{z}\right)^4 = (2 \cos \theta)^4 = 16 \cos^4 \theta$ $\therefore 16 \cos^4 \theta = 2 \cos 4\theta + 8 \cos 2\theta + 6$ $\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$	1 1 1	
6(c)(iii)	$\int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta = \int_0^{\frac{\pi}{2}} \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8} d\theta$ $= \left[\frac{1}{8} \frac{\sin 4\theta}{4} + \frac{1}{2} \frac{\sin 2\theta}{2} + \frac{3}{8} \theta \right]_0^{\frac{\pi}{2}}$ $= \frac{1}{32} \sin 2\pi + \frac{1}{4} \sin \pi + \frac{3\pi}{16} - 0$ $= \frac{3\pi}{16}$	1 1	

Question	Criteria	Marks	Bands
7(a)	Let roots be: $\alpha - d, \alpha, \alpha + d$ sum of roots: $3\alpha = -a \Rightarrow \alpha = -\frac{a}{3}$ $\alpha = -\frac{a}{3}$ is a root to: $x^3 + ax^2 + bx + c = 0$ $\left(-\frac{a}{3}\right)^3 + a\left(-\frac{a}{3}\right)^2 + b\left(-\frac{a}{3}\right) + c = 0$ $-\frac{a^3}{27} + \frac{a^3}{9} - \frac{ab}{3} + c = 0$ $-a^3 + 3a^3 - 9ab + 27c = 0$ $2a^3 - 9ab + 27c = 0$	1 1 1 1	
7(b)(i)	Consider the intersection of the two curves: $y = \frac{c}{x^2}$ $\frac{x^2}{a^2} + \frac{c^4}{x^2 b^2} = 1 \quad x^4 b^2 - x^2 a^2 b^2 + a^2 c^4 = 0$ Solving for x^2: $\Delta = a^4 b^4 - 4a^2 b^2 c^4$ for the roots to be real and distinct: $\Delta > 0$ $a^4 b^4 - 4a^2 b^2 c^4 > 0$ $a^2 b^2 > 4c^2 \quad \text{or} \quad ab > 2c^2$ If $ab > 2c^2$, x^2 has two distinct values and hence x has 4 values corresponding to 4 points of intersection.	1 1 1	
7(b)(ii)	$y = \frac{c^2}{x} \quad xa \sin \theta - \frac{c^2}{x} b \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$ $x^2 a \sin \theta - c^2 b \cos \theta = (a^2 - b^2) x \sin \theta \cos \theta$ $x^2 a \sin \theta - (a^2 - b^2) x \sin \theta \cos \theta - c^2 b \cos \theta = 0$ $\Delta = \left[(a^2 - b^2)^2 \sin \theta \cos \theta \right]^2 + 4ac^2 b \cos \theta \sin \theta$ If $0 < \theta < \frac{\pi}{2}$, $0 < \sin \theta < 1$, $0 < \cos \theta < 1$, $a, b > 0$ $\therefore \Delta > 0$ and this gives two values for x.	1 1 1	

7(b)(iii)	$x_1 + x_2 = \frac{2(a^2 - b^2)\sin\theta\cos\theta}{2a\sin\theta} = \frac{(a^2 - b^2)\sin\theta\cos\theta}{a\sin\theta}$ $\frac{x_1 + x_2}{2} = \frac{(a^2 - b^2)\sin\theta\cos\theta}{2a\sin\theta}$ $x = \frac{(a^2 - b^2)\cos\theta}{2a} \quad (1)$ sub into normal to find y: $a \frac{(a^2 - b^2)\cos\theta}{2a} \sin\theta - yb\cos\theta = (a^2 - b^2)\sin\theta\cos\theta$ $\frac{(a^2 - b^2)}{2} \sin\theta - yb = (a^2 - b^2)\sin\theta \quad \cos\theta \neq 0$ $y = \frac{-(a^2 - b^2)}{2b} \sin\theta \quad (2)$	1	
7(b)(iv)	Eliminate θ: From (1) $\cos\theta = \frac{2ax}{a^2 - b^2}$ From (2) $\sin\theta = -\frac{2by}{a^2 - b^2}$ $\sin^2\theta + \cos^2\theta = 1$ $\frac{4a^2x^2}{(a^2 - b^2)^2} + \frac{4b^2y^2}{(a^2 - b^2)^2} = 1$ $\frac{x^2}{\left(\frac{a^2 - b^2}{2a}\right)^2} + \frac{y^2}{\left(\frac{a^2 - b^2}{2b}\right)^2} = 1$	1 1 1	

Question	Criteria	Marks	Bands
8(a)(i)	Domain: $e^x > 0$ for all x for $\cos^{-1}(e^x): -1 \leq e^x \leq 1$ only if $0 \leq e^x \leq 1$, i.e. if $x \leq 0$. ☒ Range: For this domain range will be : $0 \leq y \leq \frac{\pi}{2}$ ☒	2	
8(a)(ii)	☒ limit at $y = 1$ ☒ shape 	2	
8(a)(iii)	 ☒ limit at $y = 1$ ☒ shape (reciprocal of (ii))	2	

8(b)(i)	prove true for $n = 1$ $\therefore (1 + \sqrt{2})^1 = 1 + \sqrt{2}$ true where $p_n = 1$ and $q_n = 1$ ☒ assume true for $n = k$ $\therefore (1 + \sqrt{2})^k = p_k + q_k \sqrt{2}$ prove true for $n = k + 1$ $\therefore (1 + \sqrt{2})^{k+1} = (1 + \sqrt{2})^k (1 + \sqrt{2})^1$ $= (p_k + q_k \sqrt{2})(1 + \sqrt{2})$ (by assumption above) $= p_k + p_k \sqrt{2} + q_k \sqrt{2} + 2q_k$ $= (p_k + 2q_k) + (p_k + q_k) \sqrt{2}$ ☒ since p_k and q_k are integers $\therefore p_k + 2q_k$ is an integer $= p_{k+1}$ $\therefore p_k + q_k$ is an integer $= q_{k+1}$ hence $(1 + \sqrt{2})^{k+1} = p_{k+1} + q_{k+1} \sqrt{2}$ ☒ If true for $n = k$ and $n = k + 1$ and since true for $n = 1, 2, 3, \dots$ $\therefore$ true for $\forall n$ positive integers	3	
8(b)(ii)	$p_1^2 - 2q_1^2 = 1 - 2 \times 1^2 = -1 = (-1)^1$ If $p_k^2 - 2q_k^2 = (-1)^k$ then when $n = k + 1$ $(p_{k+1})^2 - (q_{k+1})^2 = (p_k + 2q_k)^2 - 2(p_k + q_k)^2$ (from above) ☒ $= p_k^2 + 4p_k q_k + 4q_k^2 - 2p_k^2 - 4p_k q_k - 2q_k^2$ $= 2q_k^2 - p_k^2$ $= -1(p_k^2 - 2q_k^2)$ $= -1 \times (-1)^k$ $= (-1)^{k+1}$ ☒ if true for $n = k$ and $n = k + 1$ and since true for $n = 1, 2, 3, \dots$ $\therefore$ true for $\forall n$ positive integers	2	
8(c)(i)	$f(xa) = f(x) + f(a)$ if $x = 1$ then $f(a) = f(1) + f(a) \Rightarrow f(1) = 0$ if $x = a$ then $f(a^2) = f(a) + f(a)$ $f(a^2) = 2f(a)$ ☒ $\therefore 2f(-1) = f(-1^2) = f(1) = 0$ ☒	2	
8(c)(ii)	since $2f(a) = f(a^2)$ $\therefore 2f(-a) = f((-a)^2)$ $= f(a^2)$ $= 2f(a)$ ☒ since $2f(a) = 2f(-a)$ $\therefore f(a) = f(-a)$ even function ☒	2	